

VBF-T9R4-CALC-001 Calculation Record (decisive numbers)

t9_r4 - grid-storage cell cathode on the Chen2020 NMC811/graphite profile

ED feasibility (binds the family choice):

- Chen2020 positive active mass $M_{am} = 75.6e-6 \times 0.1027 \times 0.665 \times 3262 = 16.84$ g.
- calc-energy total mass = 43.45 g (porosity-corrected layers, no electrolyte).
- Baseline 1C discharge measured 4.948 Ah / 17.39 Wh -> ED 400.29 Wh/kg.
- Family capacity bounds ($C = 0.7 \times F/3.6$ / MW): layered 180-285, olivine ≤ 128 , spinel ≤ 113 , favorites ≤ 111 , NASICON ≤ 46 mAh/g.
- Non-layered ED under the 0.9 rule: olivine at 4.8 V/128 mAh/g ~ 322 Wh/kg; LiNiPO₄ at its true 5.1 V ~ 317 Wh/kg - both < 327.18 -> only layered LiMO₂ can pass (ED ~ 445 -465 Wh/kg).
- Plateau-OCP mapped discharge is anode-limited (~ 5.1 Ah, LNMO.json precedent): ED ~ 450 -500 Wh/kg for any layered at 4.6-4.8 V -> the binding constraint is the computed voltage itself.

Voltage screening (run-comp CHGNet, bcc Li reference $e_{li} = -1.87848$ eV):

- TM-redox layered band: 2.56-3.99 V (42-candidate sweep + known set; NiFe₅₅ 3.990 max).
- d0/d10 O-redox line: LiScO₂ 4.298, LiGaO₂ 4.479, LiZnO₂ 4.492, LiAlO₂ 4.598, AlB₅₅ 4.641, BGa₅₅ 4.504, GaAl₅₅ 4.494, LiBO₂ 4.253.

Charging-potential gate (comp/gate_charging.py, same machinery/seed; nested removal order):

- $V(0.4 \rightarrow 0.3) = -(E(0.4) - E(0.3) - 1 \times e_{li}) / 1$
- $E(0.4) = -295.256$ eV (gate relaxation), $E(0.3) = -285.968$ eV (batch3), $e_{li} = -1.878$ eV
- charging_potential_v = 7.410 V $\gg$ 4.8 V -> FAIL. The $x=0.3$ endpoint sits ~ 9 eV off its composition neighbors; the 50:50 avg peak was this artifact's signature.

Capacity proxy: $C = 0.7 \times 26801.481$ / MW. AlB₅₅: MW 82.57 -> 324.38 mAh/g.

Relaxed-cell density (gate full state): 2.836 g/cm³ (relaxed-final-structure).